View results

Respondent

43 Anonymous

138:54
Time to complete

A. General information about the respondant

1. * Country

Türkiye

2. * Name of organization(s) or entity(s) responsible for the preparation of the report

The Scientific and Technological Research Council of Türkiye (TÜBİTAK) National Academic Network and Information Center (ULAKBİM)

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

https://ulakbim.tubitak.gov.tr/en

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

Please describe the role/mandate of your organization: ULAKBIM's main objectives have been set as operating a high speed computer network enabling interaction within the research and education elements of the national innovation system, and providing information technology and library and information services to the research community in Türkiye. ULAKBIM aims at providing IT facilities such as computer networks, high performance compute and storage solutions, data and cloud services, and library and information services, to meet the needs of universities and research institutions, and to increase the efficiency and productivity of their users. ULAKBIM consists of National Academic Network (ULAKNET) Unit, which operations national research and education network in Türkiye, Cahit Arf Information Center, which provides information and document supply services nationwide and educational technologies department which provides open source IT solutions for middle and high schools.

7. * Full name of officially designated contact person(s) who completed this survey

Ayşe Damla ÇELİK

8. * Position of officially designated contact person(s) who completed this survey

Specialist

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

11. Brief description of the consultation process established for the preparation of the report

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

ANKOS (Anatolian University Libraries Consortium)

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey (Please use commas to separate multiple links, as: link A, link B)

https://ankos.org.tr/en/

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- [ ] Public
- [ ] Private
- [x] Civil Society
- [ ] Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- (selected) Yes
- No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Industry and Technology The Scientific and Technological Research Council of Türkiye

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- Yes
- (selected) No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- (selected) Yes
- No

20. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Establishing Open Science Policies: The Scientific and Technological Research Council of Türkiye (TÜBİTAK) has begun to promote open access to scientific publications and data by adopting the principles of open science with the "Open Science Policy" published in 2019. In 2021, TÜBİTAK prepared the draft National Open Science Strategy and Action Plan and determined the goals of disseminating open science at the national level. In projects supported by TUBITAK, researchers are required to prepare a data management plan (DMP). This policy aims to ensure that research data is more transparent and accessible. TÜBİTAK supports the transition of scientific journals to open access through the DergiPark platform. 2904 scientific journals in Türkiye are now free and open access on this platform. TR Dizin, a citation and analysis system developed by ULAKBIM, serves within the scope of open science to make scientific journals and articles in Turkey accessible. The Izmir Institute of Technology (IZTECH) played a significant role by participating in the OpenAIREPlus Project in 2011, the OpenAIRE 2020 Project in 2015, and the OpenAIRE Advance Project in 2018. From December 2011 to February 2021, these OpenAIRE projects aimed to shift scholarly communication towards openness and transparency, facilitating innovative ways to communicate and monitor research. During this period, IZTECH participated in 123 events (26 of which were international), delivered 100 presentations, and organized 15 webinars. Twelve major events were held, including the Turkish Open Science Summit 2018, which was a high-level meeting involving decision-makers and key players. This summit marked the first time all stakeholders came together, making it a landmark event for Open Science and Open Innovation in Türkiye. For more information, visit: https://libguides.iyte.edu.tr/openaire Following the OpenAIRE project, IZTECH continues to host national and international events and present papers at national and international events on open science. For exaple, within the scope of the EOSC Future Project, the "European Open Science Cloud (EOSC) Trainer Training" was held on 17-18 July 2023, hosted by IZTECH. Additionally,the euroCRIS Autumn 2025 meeting will be hosted by the IZTECH in Türkiye. Various open science-related non-governmental organizations and volunteer groups have organized events to raise public awareness and encourage open science practices. For example, communities such as Creative Commons Türkiye are actively working on this issue. TÜBİTAK ULAKBİM's agreements with Springer Nature and Wiley support open science by enabling Turkish researchers to publish their work open access in prestigious journals, increasing global visibility and accessibility of their findings. More than 160 out of 208 higher education institutions in Turkey have established Institutional Repositories and they comply with international standards. Volunteers and companies are working hard to establish Institutional Repositories in all universities. See(https://docs.google.com/spreadsheets/d/1k1UMDJPBlzOKEGk6PRYttw6lHeJ3V3nDrndLNhD6gEk/edit?gid=2107335516#gid=2107335516) The contents of the Institutional Repository in Turkey are centrally available to researchers in the Harvesting (Harman) System (https://harman.ulakbim.gov.tr/), which serves under the umbrella of TÜBİTAK - ULAKBİM. Civil society organizations in Turkey organize events every year to raise awareness and to raise awareness among wider audiences on this issue within the scope of Open Access Week. https://acikerisim.org/#

21. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

- (selected) Yes
- No

22. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

 The agreements with Springer Nature and Wiley will continue until the end of 2025, ensuring sustained support for open science initiatives. Negotiations for new agreements are underway to further expand opportunities for Turkish researchers to share their work openly and globally. Many universities pay APC fees within the scope of the Publication Incentive Scheme in order to contribute to international visibility, to increase the access of more researchers to research results, and to increase the impact value of research. Within the scope of ANKOS, there are read and publish agreements with the Association for Computing Machinery (ACM), American Chemical Society (ACS), BMJ Journals Online Collection, Cambridge Journals Online, Karger, Oxford Journals Online, Sage.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

23. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

- (selected) Yes
- No

24. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: **https://tinyurl.com/27p5zct8** designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

25. 2.1 cont. **If yes:**
    - does it include any of the following open science elements:

- [x] open access to scientific knowledge
- [x] open science infrastructure
- [ ] open engagement of societal actors
- [ ] open dialogue with other knowledge systems

26. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

    [i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ○ Yes
- ○ Partly
- ◉ No

27. 2.2 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

> ▯ The fact that issues such as open science and open access have not been sufficiently disseminated among scientists and academics in Türkiye may constitute an obstacle to the creation of a broader policy framework in this field. ▯ The infrastructure and institutions required to spread open science practices in Türkiye are still in the development stage. ▯ Investments and allocation of resources related to open science in Türkiye may not have been prioritized. ▯ Developing scientific policies sometimes requires long processes. Integrating a complex and wide-ranging issue such as open science into national policies can take time because strong cooperation and coordination between all stakeholders (academics, research institutions, government bodies, etc.) is required. ▯ Inter-institutional coordination is weak in Türkiye. It is difficult to find responsible people in institutions on any issue.

28. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

    [i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ◉ Yes
- ○ Under development
- ○ No

29. 2.3 cont. **If yes or under development,** please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

 TÜBİTAK's Open Science Policy: TÜBİTAK has developed open access and open data policies to support open science practices. In this context, open access to the outputs of projects funded by TÜBİTAK is encouraged. Higher Education Council (YÖK) Open Science Studies: YÖK has undertaken various initiatives to increase open access and open science awareness in universities. In this context, national academic archive systems (e.g. Academic Archives Platform) have been developed. Open Science Workshops and Conferences: Open science events organized by universities, TUBITAK and other organizations aim to raise awareness and disseminate good practices. Council of Higher Education Higher Education Open Access and Institutional Repository Policy: Collecting, organizing and long-term storage of all academic outputs produced at universities through Institutional Academic Archives; ensuring that all scientific outputs produced by the Turkish Higher Education System are centrally collated, quantified and made available to all national and international users as open access through the National Academic Archive system. (https://ankos.org.tr/wp-content/uploads/2018/10/kurumsal_arsiv_YOK.pdf) DRIVER Rehberi 2.0: İçerik Sağlayıcılar İçin Rehber – OAI-PMH ile Metinsel Bilgi Kaynaklarının Keşfi: By translating this reberin into Turkish, the infrastructure for standardized data entry into institutional academic archives has been prepared.

30. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

- ○ Yes
- ○ Under development
- ◉ No

31. 2.4 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

 The fact that issues such as open science and open access have not been sufficiently disseminated among scientists and academics in Türkiye may constitute an obstacle to the creation of a broader policy framework in this field. The infrastructure and institutions required to spread open science practices in Türkiye are still in the development stage. Investments and allocation of resources related to open science in Türkiye may not have been prioritized. Developing scientific policies sometimes requires long processes. Integrating a complex and wide-ranging issue such as open science into national policies can take time because strong cooperation and coordination between all stakeholders (academics, research institutions, government bodies, etc.) is required.

32. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

- ◉ Yes
- ○ Under development
- ○ No

33. 2.5 cont. **If yes,** please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).
To view the elements of open science,
see: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9

 TUBITAK has aligned its policies with the 2021 UNESCO Open Science Recommendation to promote open science.TUBITAK Open Science Policy was discussed and accepted at the meeting of the TUBITAK Board of Directors on March 14, 2019.This policy was based on the OpenAIRE model Policy on Open Science for Research Funding Organisations (RFOs). The management, storage, archiving and digital preservation of publications and research data obtained from projects conducted or supported by the Scientific and Technological Research Council of Türkiye (TUBITAK) constitute the framework of TUBITAK's Open Science Policy. Universities in Türkiye have adopted their own open access policies and open science strategies to promote open science. At the institutional level, IZTECH became the first university in Türkiye to establish a mandatory Open Access Policy in 2013 (https://hdl.handle.net/11147/7216) and an Open Science Policy (http://hdl.handle.net/11147/51) based on the OpenAIRE model for Research Performing Organizations in 2019. These policies have served as a model for other universities. In 2022, the IZTECH Research Data Management (RDM) Directory, which is Türkiye's first RDM Directory (https://hdl.handle.net/11147/12515), was approved. Many universities in Türkiye have their own open science policy.

34. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- (selected) Yes
- Under development
- No

35. 2.6 cont. **If yes or under development,** please provide details and links as relevant.

• DergiPark, operating under TÜBİTAK, provides hosting services for open-access journals. The financial support provided by TÜBİTAK contributes to the sustainability of open-access journals and the widespread dissemination of scientific knowledge. • TÜBİTAK Academic Journals play a significant role in promoting open access by providing free access to readers and not charging any fees to authors. These journals are part of a broader effort to make scientific knowledge more accessible and transparent. By fostering a culture of openness and collaboration, these initiatives, including TÜBİTAK Academic Journals, ensure that researches are widely available and can benefit both the academic community and the public. • 78 out of 208 universities in Türkiye are foundation universities. In particular, some of the foundation universities provide APC fee support for open access publications at Q1 and Q2 level to support researcher and institutional visibility.

36. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

- (selected) Yes
- No

37. 2.7 cont. **If yes**, please provide details and links as relevant.

 Integration of Open Access Agreements into the EKUAL Consortium: Through the TÜBİTAK EKUAL (National Academic License for Electronic Resources) consortium, universities in Türkiye have access to various research databases. In recent years, open access agreements have been integrated into this program, enabling researchers to publish their work openly while still benefiting from EKUAL's extensive database support.

38. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

- (selected) Yes
- No

39. 2.8 cont. **If yes**, please provide details and links as relevant.

 TÜBİTAK actively encourages partnerships between academia, industry, and public institutions to promote open science practices. Research outputs from collaborative projects supported by ARDEB (Research Support Programs Directorate) are made accessible through open access platforms like Aperta. Future plans aim to formalize open science requirements in these partnerships further. Expansion of EKUAL to Include Open Access Publishing Agreements: The EKUAL consortium, traditionally focused on academic access to research databases, is negotiating with publishers to incorporate open access publishing agreements into its offerings. These agreements aim to enable researchers from universities and industry partners to publish their work openly while benefiting from EKUAL's extensive database support. Such negotiations highlight a commitment to equitable knowledge sharing and foster closer collaboration between public and private sectors. National Open Science Strategy Development (Planned): A National Open Science Strategy, expected to be finalized by 2025, is being developed to include guidelines for fostering public-private partnerships. This strategy will emphasize equitable access, shared responsibilities, and incentives for private entities participating in open science practices.

40. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ○ Yes
- ○ Under development
- ● No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

41. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

The most up-to-date official data on the ratio of Research and Development (R&D) expenditures to Gross Domestic Product (GDP) in Türkiye is for 2023 and was published on November 6, 2024. According to this data, total R&D expenditures in 2023 were 377 billion 542 million TL and the share of R&D expenditures in GDP was 1.42%. (https://tubitak.gov.tr/tr/haber/2023-yili-ar-ge-harcamasi-377-milyar-542-milyon-tl-olarak-gerceklesti) In the previous year, 2022, R&D expenditures were recorded as 198 billion 670 million TL. These data were compiled and published by the Turkish Statistical Institute (TUIK). (https://data.tuik.gov.tr/Bulten/Index?p=Arastirma-Gelistirme-Faaliyetleri-Arastirmasi-2022-49408&utm_source)

42. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

The most recent official data on the percentage of the Turkish population with access to reliable internet and bandwidth comes from 2023 and was published in 2024 by the Turkish Statistical Institute (TÜİK). Reliable Internet Access: • In 2023, 95.3% of households in Türkiye had internet access, with 93.2% of these households using the internet daily or weekly. Bandwidth Access: • 85.1% of households had a fixed broadband internet connection. Source: • Turkish Statistical Institute (TÜİK), "Household Information Technology Usage Survey, 2023."

43. 3.3 Are there national research and education networks (NRENs)[i] active in the country? *(ref.: (iii) 18.c)*

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- ○ Yes
- ● No

44. 3.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

45. 3.4 Are there national or institutional open science infrastructures[i] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- (●) Yes
- ( ) Under development
- ( ) No

46. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- [x] federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- [ ] community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- [ ] platforms for exchanges and co-creation of knowledge between scientists and society
- [ ] community-based monitoring and information systems to complement national, regional and global data and information systems

47. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

> ▯ APERTA is a national platform that hosts and provides access to academic publications, research data, and other scientific materials in Türkiye. Managed by TÜBİTAK ULAKBİM, APERTA is designed to facilitate open access to Turkish scientific outputs, ensuring that research data and publications are freely available to the global scientific community. It serves as a repository for research results from various disciplines and promotes the adoption of open data and open access publishing practices. ▯ HARMAN is Türkiye's national research data repository, managed by TÜBİTAK. It is designed to collect, store, and provide access to research data produced by researchers and institutions across Türkiye. HARMAN supports open science practices by providing a centralized platform for the sharing and dissemination of research data, making it openly accessible to the academic community for further use, analysis, and innovation.Institutional Academic Archives: These are archives where academic institutions store their own data. There are more than 160 institutional repositories in Türkiye that comply with international standards. DergiPark: Under the umbrella of TÜBİTAK-ULAKBİM, it provides electronic hosting and editorial process management services for peer-reviewed academic journals published in Türkiye.

48. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- (●) Yes
- ( ) No

49. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open sciencei, stage of research, accessibility to all the relevant users in the region or internationally).

> ▯ TÜBİTAK ULAKBİM plays an active role in EOSC, which is a federated European infrastructure aimed at providing seamless access to research data, services, and resources across Europe. As part of Türkiye's involvement in EOSC, TÜBİTAK ULAKBİM participates in initiatives and projects that promote open access to research data and services, fostering international collaboration and data sharing across borders. ▯ Türkiye contributes to SCOAP^3 through TÜBİTAK and its funding mechanisms, supporting the open access transition in high-energy physics research. This initiative is a key example of Türkiye's participation in federated international infrastructures that promote open science, particularly in specialized fields like particle physics. ▯ OpenAIRE Türkiye Help Desk is officially carried out by IZTECH Library.

50. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

- ○ Yes
- ◉ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

51. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

- ◉ Yes
- ○ No

52. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

• Türkiye has been aligning itself with global trends in open science, including the European Union's Open Science Policy. The Turkish Council of Higher Education (YÖK) and the Scientific and Technological Research Council of Türkiye (TÜBİTAK) have shown interest in promoting open science principles. • TÜBİTAK has supported open access to research outputs and has encouraged researchers to publish in open-access journals or deposit their work in institutional repository. • Türkiye has been aligning itself with global trends in open science, including the European Union's Open Science Policy. The Turkish Council of Higher Education (YÖK) and the Scientific and Technological Research Council of Türkiye (TÜBİTAK) have shown interest in promoting open science principles. • TÜBİTAK has supported open access to research outputs and has encouraged researchers to publish in open-access journals or deposit their work in institutional repositor • Türkiye is expected to continue integrating open science into its national research and innovation strategies. This includes potential policy developments, increased funding for open science infrastructure, and expanded training programs. • The adoption of the UNESCO Recommendation on Open Science (2021) may further accelerate efforts to institutionalize open science practices in Türkiye.

53. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

- ○ Yes
- ○ No

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

54. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

- ◉ Yes
- ○ No

55. 5.1 cont. **If yes**, please provide more information and links as relevant.

56. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

- ○ Yes
- ◉ No

57. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

58. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

- ◉ Yes
- ○ No

59. 5.3 cont. **If yes,** for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

• TÜBİTAK-supported projects often require researchers to make their publications openly accessible, though this is not yet a mandatory criterion for all funding programs. • TÜBİTAK has also been involved in international open science initiatives, such as OpenAIRE, which promotes open access and open data practices. • By 2025, Türkiye is expected to develop a more comprehensive national policy framework for open science, potentially including guidelines to recognize and reward open science practices. • TÜBİTAK and YÖK could introduce specific criteria for evaluating open science contributions, such as data sharing, open access publishing, and societal impact, in academic promotions and funding decisions. • By 2025, Türkiye could align more closely with global standards for recognizing and rewarding open science practices. • IZTECH is involved in OpenAIRE projects and serves as the OpenAIRE Türkiye NOAD (National Open Access Desk). Additionally, it actively participates in various working groups, including the OpenAIRE RDM Working Group and the OpenAIRE Training Working Group. Furthermore, IZTECH is a member of the COAR Repository Assessment Working Group, the COAR Controlled Vocabularies Working Group, and the COAR Task Force on Managing Multilingual and Non-English Content.

60. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

- ○ Yes
- ○ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

61. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

- ○ YES, at the national level
- ○ YES, at the institutional level
- ○ YES, at both the national and institutional levels
- ○ No

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

62. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

 Türkiye has been aligning its national research and innovation policies with international open science frameworks, such as the European Union's Open Science Policy and the UNESCO Recommendation on Open Science. TÜBİTAK (The Scientific and Technological Research Council of Türkiye) and the Council of Higher Education (YÖK) have been key players in promoting open science principles, including open access, open data, and international collaboration. Türkiye is an active participant in international open science initiatives, such as: OpenAIRE: A European network that promotes open access to research outputs and open science practices. COAR (Confederation of Open Access Repositories): A global association that supports the development of open access repositories and open science infrastructure. Türkiye's participation in Horizon 2020 and Horizon Europe has been a major driver of open science practices. These programs require funded projects to adhere to open science principles, such as open access publishing, open data sharing, and public engagement.

63. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

- ○ Yes
- ◉ No

64. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

65. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

- ◉ Yes
- ○ No

66. 7.3 cont. **If yes**, please provide links as relevant and more information.

 OpenAIRE: A European network that promotes open access to research outputs and open science practices. COAR (Confederation of Open Access Repositories): A global association that supports the development of open access repositories and open science infrastructure. Türkiye's participation in Horizon 2020 and Horizon Europe has been a major driver of open science practices. These programs require funded projects to adhere to open science principles, such as open access publishing, open data sharing, and public engagement

Part 8. GENERAL CONSIDERATIONS

67. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

- ○ Yes
- ◉ No

68. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org